



A Web Platform for Tensor Decomposition, Comparison, and Analysis

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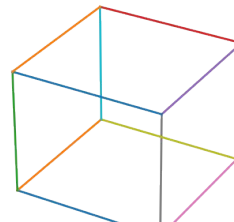


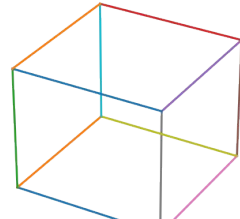


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Problem Statement

- Increasing use of high-dimensional tensor data in modern applications.
- High computational and storage costs associated with tensor operations.
- Inability of traditional matrix decomposition methods to preserve multiway tensor relationships.





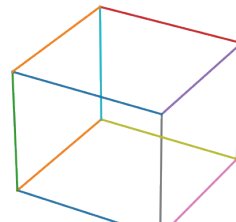
Problem Statement

- Complexity and lack of accessibility of tensor decomposition techniques.
- Existing tensor decomposition tools are fragmented across libraries and difficult to compare.
- Lack of integrated benchmarking and performance analysis systems for decomposition methods.



Proposed Solution (Tensor Decomposer)

- Tensor input through web interface
- Tensor decomposition using multiple methods
- Reconstruction of tensors from decomposed matrices and accuracy analysis





Proposed Solution

- Compression ratio analysis
- Benchmarking of decomposition method and execution time
- Visualization of decomposed outputs
- Comparison between methods



Decomposition Methods

- CP Decomposition
- Tucker Decomposition
- HOSVD
- Tensor Train Decomposition



Key Technologies & Tools

- **Language:** Python, Javascript
- **GPU Support:** PyTorch
- **Data Processing & Reporting:** Pandas
- **Visualization:** Matplotlib
- **Version control:** Git, Github

Thank you